Slope stability analysis

Input data (Construction stage 1 - End of Construction)

Project

Project :
Part : P2. TRDSRB STA. 3+750_(BH-86)
Description : SLOPE STABILITY OF HIGHEST EMBANKMENT
Date : 8/7/2026

Settings

Philippines - DPWH Settings

Stability analysis

Verification methodology : Safety factors (ASD)
Earthquake analysis : Standard

Safety factors			
Permanent design situation			
Safety factor :	SF_s =	1.50	[–]

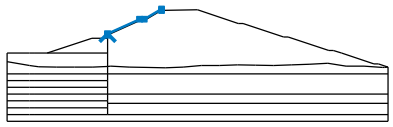
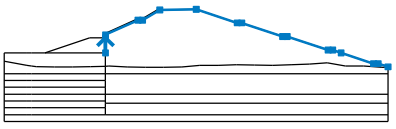
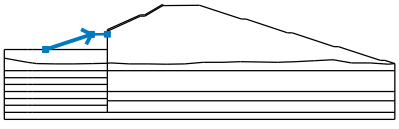
Safety factors			
Seismic design situation			
Safety factor :	SF_s =	1.10	[–]

Anchors

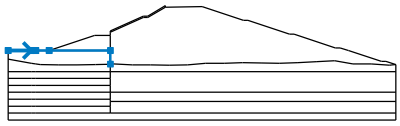
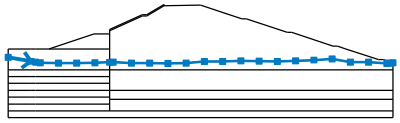
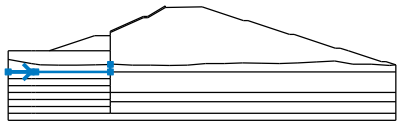
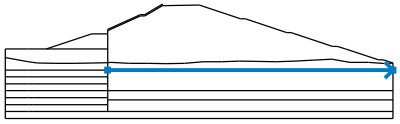
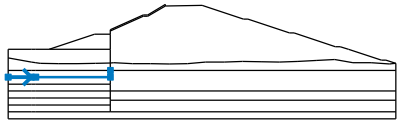
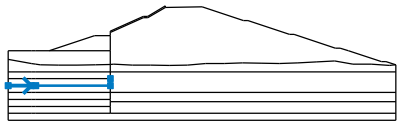
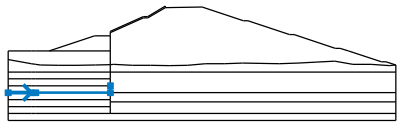
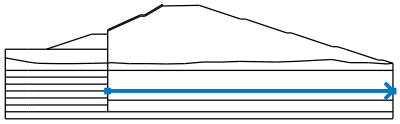
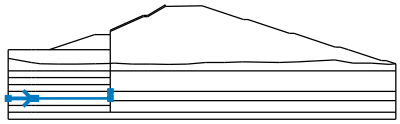
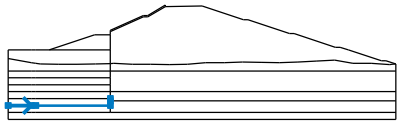
Verification methodology : Safety factors (ASD)

Safety factors			
Safety factor for steel strength :	SF_t =	1.50	[–]
Safety factor for pull out resistance (soil) :	SF_e =	1.50	[–]
Safety factor for pull out resistance (grouting) :	SF_c =	1.50	[–]

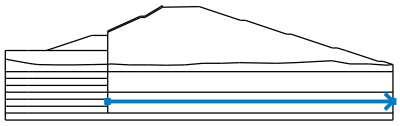
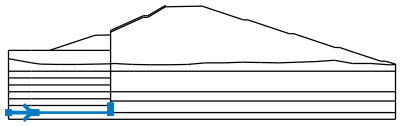
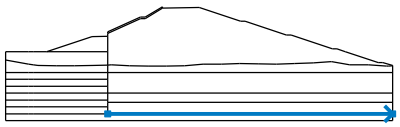
Interface

No.	Interface location	Coordinates of interface points [m]					
		x	z	x	z	x	z
1		47.25	0.78	47.25	1.08	54.37	4.30
		55.37	4.30	59.19	6.55	59.19	6.25
2		47.25	-3.19	47.25	0.10	47.25	0.78
		54.37	4.00	55.37	4.00	59.19	6.25
		67.19	6.38	76.19	3.38	77.19	3.38
		86.19	0.40	87.19	0.40	96.19	-2.58
		97.19	-2.58	99.03	-3.19	106.19	-5.56
		107.19	-5.56	109.33	-6.25		
3		33.90	-3.19	43.80	0.08	47.25	0.10

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No.	Interface location	Coordinates of interface points [m]					
		x	z	x	z	x	z
4		25.00	-3.19	30.00	-3.19	31.00	-3.19
		33.90	-3.19	47.25	-3.19	47.25	-6.13
5		25.00	-5.06	30.00	-5.92	31.00	-6.08
		32.00	-6.24	36.00	-6.31	40.00	-6.31
		44.00	-6.25	47.25	-6.13	48.00	-6.10
		52.00	-6.29	56.00	-6.33	60.00	-6.39
		64.00	-6.31	68.00	-5.95	72.00	-5.95
		76.00	-5.79	80.00	-5.86	84.00	-5.96
		88.00	-5.81	92.00	-5.64	96.00	-5.40
		100.00	-6.08	104.00	-6.10	108.00	-6.38
		108.51	-6.33	109.33	-6.26		
6		25.00	-7.75	30.00	-7.75	31.00	-7.75
		47.25	-7.75	47.25	-6.13		
7		47.25	-7.75	109.33	-7.75		
8		25.00	-9.25	30.00	-9.25	31.00	-9.25
		47.25	-9.25	47.25	-7.75		
9		25.00	-10.75	30.00	-10.75	31.00	-10.75
		47.25	-10.75	47.25	-9.25		
10		25.00	-12.25	30.00	-12.25	31.00	-12.25
		47.25	-12.25	47.25	-10.75		
11		47.25	-12.25	109.33	-12.25		
12		25.00	-13.75	30.00	-13.75	31.00	-13.75
		47.25	-13.75	47.25	-12.25		
13		25.00	-15.25	30.00	-15.25	31.00	-15.25
		47.25	-15.25	47.25	-14.25	47.25	-13.75

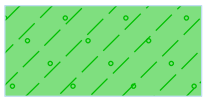
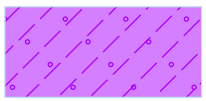
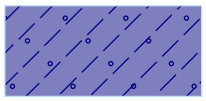
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No.	Interface location	Coordinates of interface points [m]					
		x	z	x	z	x	z
14		47.25	-14.25	109.33	-14.25		
15		25.00	-16.75	30.00	-16.75	31.00	-16.75
		47.25	-16.75	47.25	-15.25		
16		47.25	-16.75	109.33	-16.75		

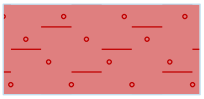
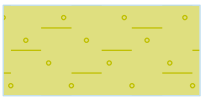
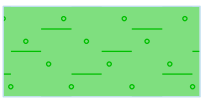
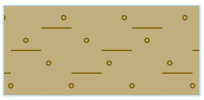
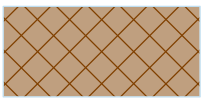
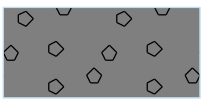
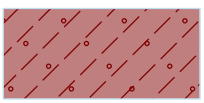
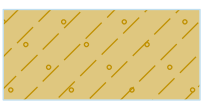
Soil parameters - effective stress state

No.	Name	Pattern	φ_{ef} [°]	c_{ef} [kPa]	γ [kN/m^3]
1	SM1- Silt		35.00	10.00	19.00
2	MH1- Silty Clay		5.00	79.39	17.00
3	MH2- Silty Clay		5.00	69.53	17.00
4	SM2- Silt		7.89	0.00	14.36
5	Embankment		35.00	10.00	20.00
6	Foot Protection		41.00	1.00	25.00
7	EXISTING_MH_1.5m		0.00	70.00	19.94
8	EXISTING_SM_3m		27.00	0.00	17.94
9	EXISTING_MH_4.5m		0.00	24.00	18.33

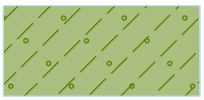
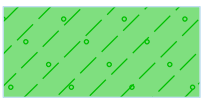
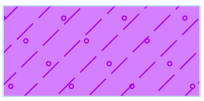
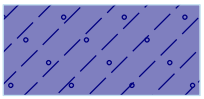
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No.	Name	Pattern	φ_{ef} [°]	c_{ef} [kPa]	γ [kN/m^3]
10	EXISTING_SM_6m		17.00	0.00	14.92
11	EXISTING_SM_7.5m		12.00	0.00	15.71
12	EXISTING_SM_9m		12.00	0.00	14.14
13	EXISTING_SM_10.5m		8.00	0.00	15.71
14	EXISTING_SM_12m		3.00	0.00	13.35

Soil parameters - uplift

No.	Name	Pattern	γ_{sat} [kN/m^3]	γ_s [kN/m^3]	n [–]
1	SM1- Silt		19.00		
2	MH1- Silty Clay		17.00		
3	MH2- Silty Clay		17.00		
4	SM2- Silt		16.92		
5	Embankment		20.00		
6	Foot Protection		25.00		
7	EXISTING_MH_1.5m		19.94		
8	EXISTING_SM_3m		18.36		

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No.	Name	Pattern	γ_{sat} [kN/m3]	γ_s [kN/m3]	n [–]
9	EXISTING_MH_4.5m		18.33		
10	EXISTING_SM_6m		17.28		
11	EXISTING_SM_7.5m		17.63		
12	EXISTING_SM_9m		16.89		
13	EXISTING_SM_10.5m		17.48		
14	EXISTING_SM_12m		16.29		

Soil parameters

SM1- Silt

Unit weight : $\gamma = 19.00$ kN/m3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : $\varphi_{ef} = 35.00$ °
Cohesion of soil : $c_{ef} = 10.00$ kPa
Saturated unit weight : $\gamma_{sat} = 19.00$ kN/m3
Coeff. of reduction of pore pressure : $X = 0.10$

MH1- Silty Clay

Unit weight : $\gamma = 17.00$ kN/m3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : $\varphi_{ef} = 5.00$ °
Cohesion of soil : $c_{ef} = 79.39$ kPa
Saturated unit weight : $\gamma_{sat} = 17.00$ kN/m3
Coeff. of reduction of pore pressure : $X = 0.10$

MH2- Silty Clay

Unit weight : $\gamma = 17.00$ kN/m3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : $\varphi_{ef} = 5.00$ °
Cohesion of soil : $c_{ef} = 69.53$ kPa
Saturated unit weight : $\gamma_{sat} = 17.00$ kN/m3
Coeff. of reduction of pore pressure : $X = 0.10$

SM2- Silt

Unit weight : $\gamma = 14.36$ kN/m3

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Stress-state :	effective
Shear strength :	Mohr-Coulomb
Angle of internal friction :	φ_{ef} = 7.89 °
Cohesion of soil :	c_{ef} = 0.00 kPa
Saturated unit weight :	γ_{sat} = 16.92 kN/m^3
Coeff. of reduction of pore pressure :	X = 0.10

Embankment

Unit weight :	γ = 20.00 kN/m^3
Stress-state :	effective
Shear strength :	Mohr-Coulomb
Angle of internal friction :	φ_{ef} = 35.00 °
Cohesion of soil :	c_{ef} = 10.00 kPa
Saturated unit weight :	γ_{sat} = 20.00 kN/m^3
Coeff. of reduction of pore pressure :	X = 1.00

Foot Protection

Unit weight :	γ = 25.00 kN/m^3
Stress-state :	effective
Shear strength :	Mohr-Coulomb
Angle of internal friction :	φ_{ef} = 41.00 °
Cohesion of soil :	c_{ef} = 1.00 kPa
Saturated unit weight :	γ_{sat} = 25.00 kN/m^3
Coeff. of reduction of pore pressure :	X = 1.00

EXISTING_MH_1.5m

Unit weight :	γ = 19.94 kN/m^3
Stress-state :	effective
Shear strength :	Mohr-Coulomb
Angle of internal friction :	φ_{ef} = 0.00 °
Cohesion of soil :	c_{ef} = 70.00 kPa
Saturated unit weight :	γ_{sat} = 19.94 kN/m^3
Coeff. of reduction of pore pressure :	X = 0.10

EXISTING_SM_3m

Unit weight :	γ = 17.94 kN/m^3
Stress-state :	effective
Shear strength :	Mohr-Coulomb
Angle of internal friction :	φ_{ef} = 27.00 °
Cohesion of soil :	c_{ef} = 0.00 kPa
Saturated unit weight :	γ_{sat} = 18.36 kN/m^3
Coeff. of reduction of pore pressure :	X = 0.10

EXISTING_MH_4.5m

Unit weight :	γ = 18.33 kN/m^3
Stress-state :	effective
Shear strength :	Mohr-Coulomb
Angle of internal friction :	φ_{ef} = 0.00 °
Cohesion of soil :	c_{ef} = 24.00 kPa
Saturated unit weight :	γ_{sat} = 18.33 kN/m^3
Coeff. of reduction of pore pressure :	X = 0.10

EXISTING_SM_6m

Unit weight :	γ = 14.92 kN/m^3
Stress-state :	effective

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Shear strength : Mohr-Coulomb
Angle of internal friction : φ_{ef} = 17.00 °
Cohesion of soil : c_{ef} = 0.00 kPa
Saturated unit weight : γ_{sat} = 17.28 kN/m^3
Coeff. of reduction of pore pressure : X = 0.10

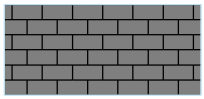
EXISTING_SM_7.5m
Unit weight : γ = 15.71 kN/m^3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : φ_{ef} = 12.00 °
Cohesion of soil : c_{ef} = 0.00 kPa
Saturated unit weight : γ_{sat} = 17.63 kN/m^3
Coeff. of reduction of pore pressure : X = 0.10

EXISTING_SM_9m
Unit weight : γ = 14.14 kN/m^3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : φ_{ef} = 12.00 °
Cohesion of soil : c_{ef} = 0.00 kPa
Saturated unit weight : γ_{sat} = 16.89 kN/m^3
Coeff. of reduction of pore pressure : X = 0.10

EXISTING_SM_10.5m
Unit weight : γ = 15.71 kN/m^3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : φ_{ef} = 8.00 °
Cohesion of soil : c_{ef} = 0.00 kPa
Saturated unit weight : γ_{sat} = 17.48 kN/m^3
Coeff. of reduction of pore pressure : X = 0.10

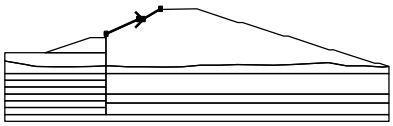
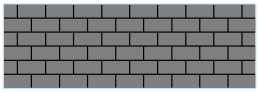
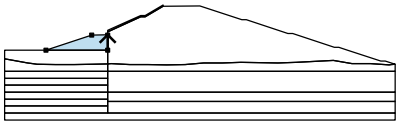
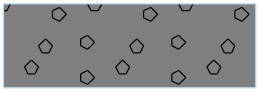
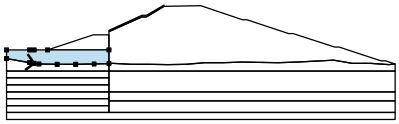
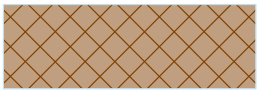
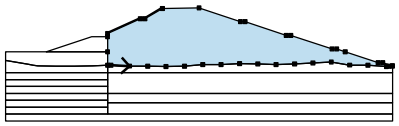
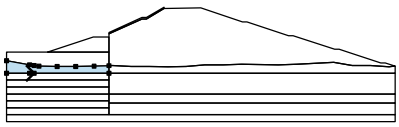
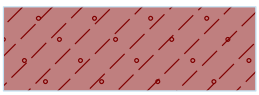
EXISTING_SM_12m
Unit weight : γ = 13.35 kN/m^3
Stress-state : effective
Shear strength : Mohr-Coulomb
Angle of internal friction : φ_{ef} = 3.00 °
Cohesion of soil : c_{ef} = 0.00 kPa
Saturated unit weight : γ_{sat} = 16.29 kN/m^3
Coeff. of reduction of pore pressure : X = 0.10

Rigid Bodies

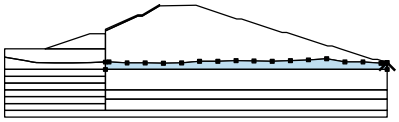
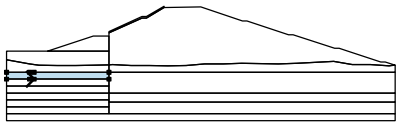
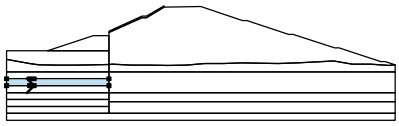
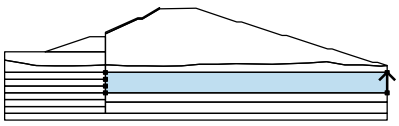
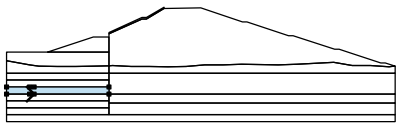
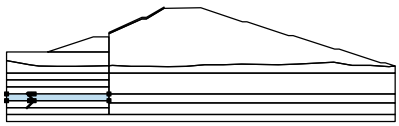
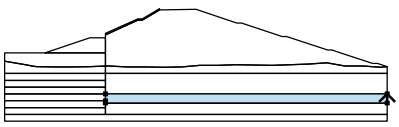
No.	Name	Sample	γ [kN/m^3]
1	Concrete Revetment		25.00

Assigning and surfaces

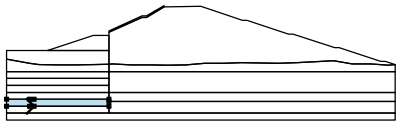
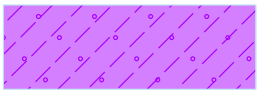
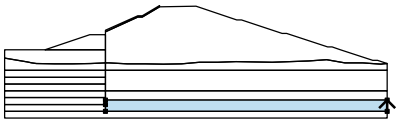
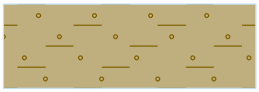
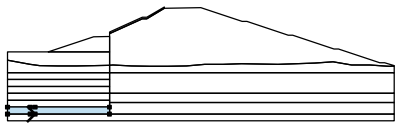
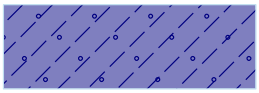
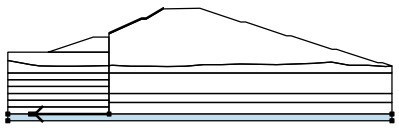
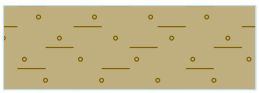
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No.	Surface position	Coordinates of surface points [m]				Assigned soil
		x	z	x	z	
1		54.37	4.00	55.37	4.00	Concrete Revetment
		59.19	6.25	59.19	6.55	
		55.37	4.30	54.37	4.30	
		47.25	1.08	47.25	0.78	
2		47.25	-3.19	47.25	0.10	Foot Protection
		43.80	0.08	33.90	-3.19	
3		30.00	-5.92	31.00	-6.08	Embankment
		32.00	-6.24	36.00	-6.31	
		40.00	-6.31	44.00	-6.25	
		47.25	-6.13	47.25	-3.19	
		33.90	-3.19	31.00	-3.19	
		30.00	-3.19	25.00	-3.19	
		25.00	-5.06			
4		48.00	-6.10	52.00	-6.29	Embankment
		56.00	-6.33	60.00	-6.39	
		64.00	-6.31	68.00	-5.95	
		72.00	-5.95	76.00	-5.79	
		80.00	-5.86	84.00	-5.96	
		88.00	-5.81	92.00	-5.64	
		96.00	-5.40	100.00	-6.08	
		104.00	-6.10	108.00	-6.38	
		108.51	-6.33	109.33	-6.26	
		109.33	-6.25	107.19	-5.56	
		106.19	-5.56	99.03	-3.19	
		97.19	-2.58	96.19	-2.58	
		87.19	0.40	86.19	0.40	
		77.19	3.38	76.19	3.38	
		67.19	6.38	59.19	6.25	
		55.37	4.00	54.37	4.00	
		47.25	0.78	47.25	0.10	
		47.25	-3.19	47.25	-6.13	
5		30.00	-7.75	31.00	-7.75	EXISTING_MH_1.5m
		47.25	-7.75	47.25	-6.13	
		44.00	-6.25	40.00	-6.31	
		36.00	-6.31	32.00	-6.24	
		31.00	-6.08	30.00	-5.92	
		25.00	-5.06	25.00	-7.75	

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No.	Surface position	Coordinates of surface points [m]				Assigned soil
		x	z	x	z	
6		109.33	-7.75	109.33	-6.26	SM1- Silt
		108.51	-6.33	108.00	-6.38	
		104.00	-6.10	100.00	-6.08	
		96.00	-5.40	92.00	-5.64	
		88.00	-5.81	84.00	-5.96	
		80.00	-5.86	76.00	-5.79	
		72.00	-5.95	68.00	-5.95	
		64.00	-6.31	60.00	-6.39	
		56.00	-6.33	52.00	-6.29	
		48.00	-6.10	47.25	-6.13	
		47.25	-7.75			
7		30.00	-9.25	31.00	-9.25	EXISTING_SM_3m
		47.25	-9.25	47.25	-7.75	
		31.00	-7.75	30.00	-7.75	
		25.00	-7.75	25.00	-9.25	
8		30.00	-10.75	31.00	-10.75	EXISTING_MH_4.5m
		47.25	-10.75	47.25	-9.25	
		31.00	-9.25	30.00	-9.25	
		25.00	-9.25	25.00	-10.75	
9		109.33	-12.25	109.33	-7.75	MH1- Silty Clay
		47.25	-7.75	47.25	-9.25	
		47.25	-10.75	47.25	-12.25	
10		30.00	-12.25	31.00	-12.25	EXISTING_SM_6m
		47.25	-12.25	47.25	-10.75	
		31.00	-10.75	30.00	-10.75	
		25.00	-10.75	25.00	-12.25	
11		30.00	-13.75	31.00	-13.75	EXISTING_SM_7.5m
		47.25	-13.75	47.25	-12.25	
		31.00	-12.25	30.00	-12.25	
		25.00	-12.25	25.00	-13.75	
12		109.33	-14.25	109.33	-12.25	MH2- Silty Clay
		47.25	-12.25	47.25	-13.75	
		47.25	-14.25			

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No.	Surface position	Coordinates of surface points [m]				Assigned soil
		x	z	x	z	
13		30.00	-15.25	31.00	-15.25	EXISTING_SM_9m
		47.25	-15.25	47.25	-14.25	
		47.25	-13.75	31.00	-13.75	
		30.00	-13.75	25.00	-13.75	
		25.00	-15.25			
14		109.33	-16.75	109.33	-14.25	SM2- Silt
		47.25	-14.25	47.25	-15.25	
		47.25	-16.75			
15		30.00	-16.75	31.00	-16.75	EXISTING_SM_10.5m
		47.25	-16.75	47.25	-15.25	
		31.00	-15.25	30.00	-15.25	
		25.00	-15.25	25.00	-16.75	
16		47.25	-16.75	31.00	-16.75	SM2- Silt
		30.00	-16.75	25.00	-16.75	
		25.00	-18.25	109.33	-18.25	
		109.33	-16.75			

Anti-Slide piles

No.	Point		Length	Construction type	Depth of beam	Length of beam	Pile spacing	
	x [m]	z [m]	l [m]		h [m]	l_b [m]	b_f [m]	b/b_b [m]
1	47.25	1.08	18.00	standard wall				0.30

No.	Cross-section [m]	Pile bearing capacity			
		Distribution along the pile	Max. bearing capacity V_u [kN]	Gradient K [–]	Passive force direction
1	s_x = 0.30; s_y = 0.30	linear	100.00	0.50	perpendicular to pile

Surcharge

No.	Type	Type of action	Location z [m]	Origin x [m]	Length l [m]	Width b [m]	Slope α [°]	Magnitude		
								q, q_1, f, F, x	q_2, z	unit
1	strip	permanent	z = 6.25	x = 59.19	l = 8.00		1.15	12.00		kN/m^2

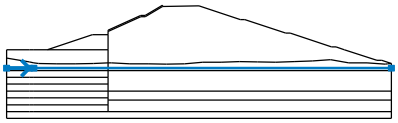
Surcharges

No.	Name
1	HL

Water

Water type : GWT

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No.	GWT location	Coordinates of GWT points [m]					
		x	z	x	z	x	z
1		25.00	-7.25	30.00	-7.25	31.00	-7.25
		109.33	-7.25				

Earthquake

Earthquake not included.

Settings of the stage of construction

Design situation : permanent

Results (Construction stage 1 - End of Construction)

Analysis 1 (stage 1)

Circular slip surface

Slip surface parameters							
Center :	x =	47.78	[m]	Angles :	α_1 =	-69.89	[°]
	z =	5.08	[m]		α_2 =	89.41	[°]
Radius :	R =	24.05	[m]				
The slip surface after optimization.							

Total weight of soil above the slip surface: 13963.46 kN/m

Slope stability verification (Bishop)

Sum of active forces : F_a = 1848.50 kN/m

Sum of passive forces : F_p = 3380.22 kN/m

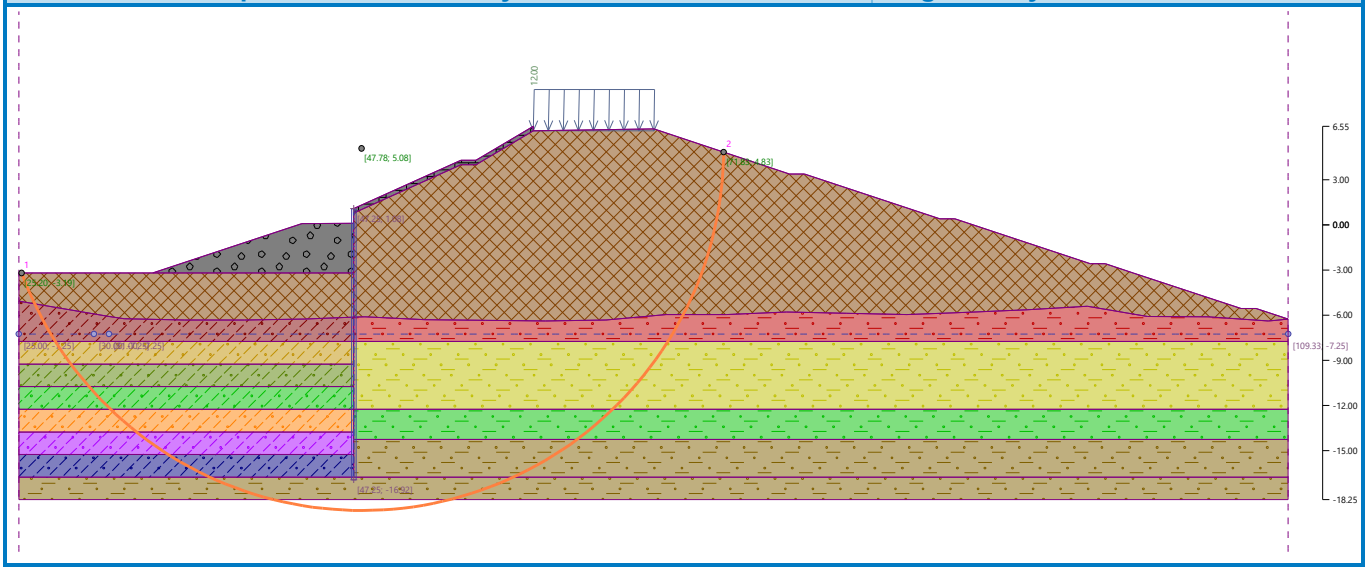
Sliding moment : M_a = 44456.45 kNm/m

Resisting moment : M_p = 81294.30 kNm/m

Factor of safety = 1.83 > 1.50

Slope stability ACCEPTABLE

Name : Global Slope Stabilization Analysis	Stage - analysis : 1 - 1



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Input data (Construction stage 2 - Steady Seepage)

Water

Water type : Pore pressure

No.	Pore pressure interface location	Coordinates of pore pressure interface points [m]						Pore press. [kPa]
		x	z	x	z	x	z	
1		25.00	4.73	30.00	4.73	31.00	4.73	108.597
		109.33	4.73					

Earthquake

Earthquake not included.

Settings of the stage of construction

Design situation : permanent

Results (Construction stage 2 - Steady Seepage)

Analysis 1 (stage 2)

Circular slip surface

Slip surface parameters							
Center :	x =	47.85	[m]	Angles :	α_1 =	-65.66	[°]
	z =	7.05	[m]		α_2 =	84.31	[°]
Radius :	R =	24.85	[m]				
The slip surface after optimization.							

Total weight of soil above the slip surface: 13334.08 kN/m

Slope stability verification (Bishop)

Sum of active forces : F_a = 1874.37 kN/m

Sum of passive forces : F_p = 3947.18 kN/m

Sliding moment : M_a = 46578.20 kNm/m

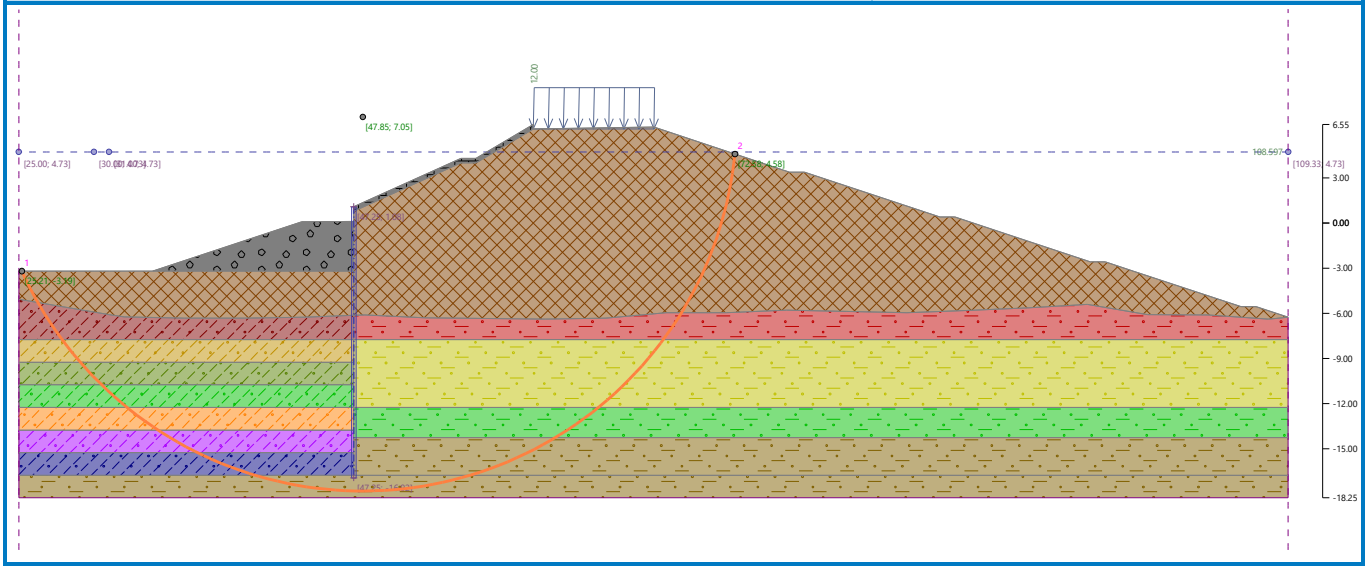
Resisting moment : M_p = 98087.44 kNm/m

Factor of safety = 2.11 > 1.50

Slope stability ACCEPTABLE

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Name : Global Slope Stabilization Analysis	Stage - analysis : 2 - 1

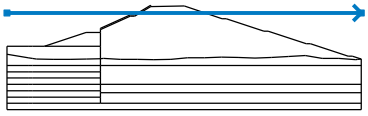
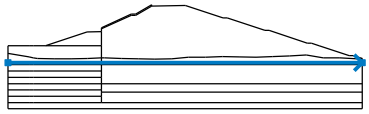


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Input data (Construction stage 3 - Rapid Drawdown)

Water

Water type : Rapid drawdown

Interface	Surface location	Coordinates of surface points [m]					
		x	z	x	z	x	z
Original GWT		25.00	4.73	109.33	4.73		
GWT		25.00	-7.25	109.33	-7.25		

Earthquake

Earthquake not included.

Settings of the stage of construction

Design situation : permanent

Results (Construction stage 3 - Rapid Drawdown)

Analysis 1 (stage 3)

Circular slip surface

Slip surface parameters							
Center :	x =	48.62	[m]	Angles :	α_1 =	-66.92	[°]
	z =	6.79	[m]		α_2 =	84.00	[°]
Radius :	R =	25.46	[m]				
The slip surface after optimization.							

Total weight of soil above the slip surface: 14379.98 kN/m

Slope stability verification (Bishop)

Sum of active forces : F_a = 1768.09 kN/m

Sum of passive forces : F_p = 2790.25 kN/m

Sliding moment : M_a = 45015.64 kNm/m

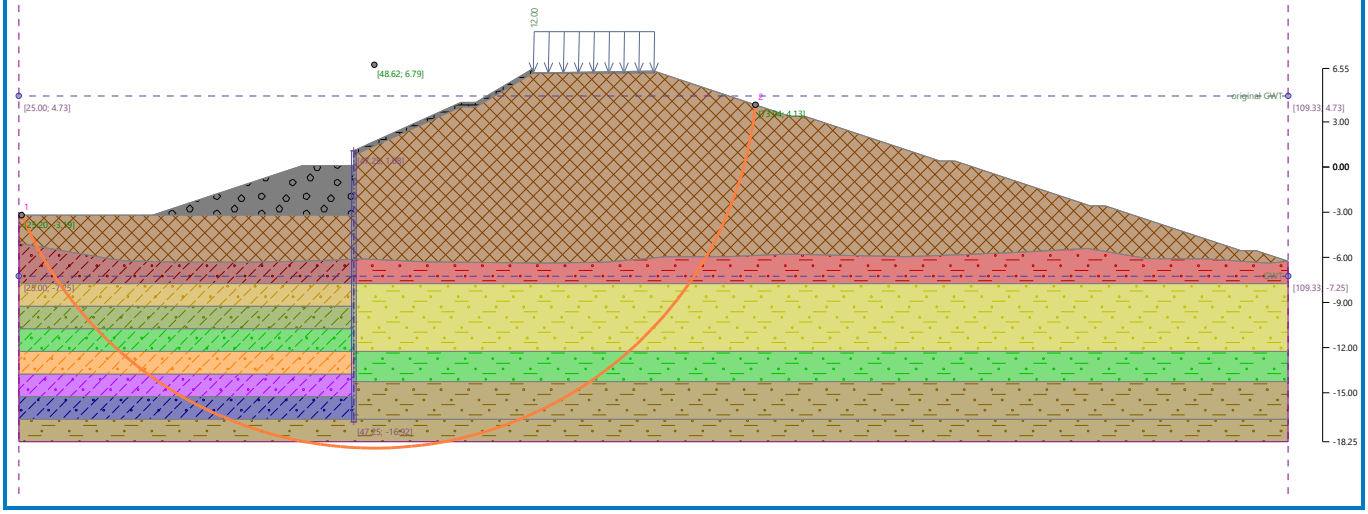
Resisting moment : M_p = 71039.79 kNm/m

Factor of safety = 1.58 > 1.50

Slope stability ACCEPTABLE

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Name : Global Slope Stabilization Analysis	Stage - analysis : 3 - 1



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Input data (Construction stage 4 - Seismic)

Water

Water type : Pore pressure

No.	Pore pressure interface location	Coordinates of pore pressure interface points [m]						Pore press. [kPa]
		x	z	x	z	x	z	
1		25.00	1.00	109.33	1.00			9.810

Earthquake

Horizontal seismic coefficient : K_h = 0.1800
Vertical seismic coefficient : K_v = 0.0000

Settings of the stage of construction

Design situation : seismic

Results (Construction stage 4 - Seismic)

Analysis 1 (stage 4)

Circular slip surface

Slip surface parameters							
Center :	x =	49.95	[m]	Angles :	α_1 =	-61.18	[°]
	z =	10.43	[m]		α_2 =	75.47	[°]
Radius :	R =	28.25	[m]				
The slip surface after optimization.							

Total weight of soil above the slip surface: 14702.55 kN/m

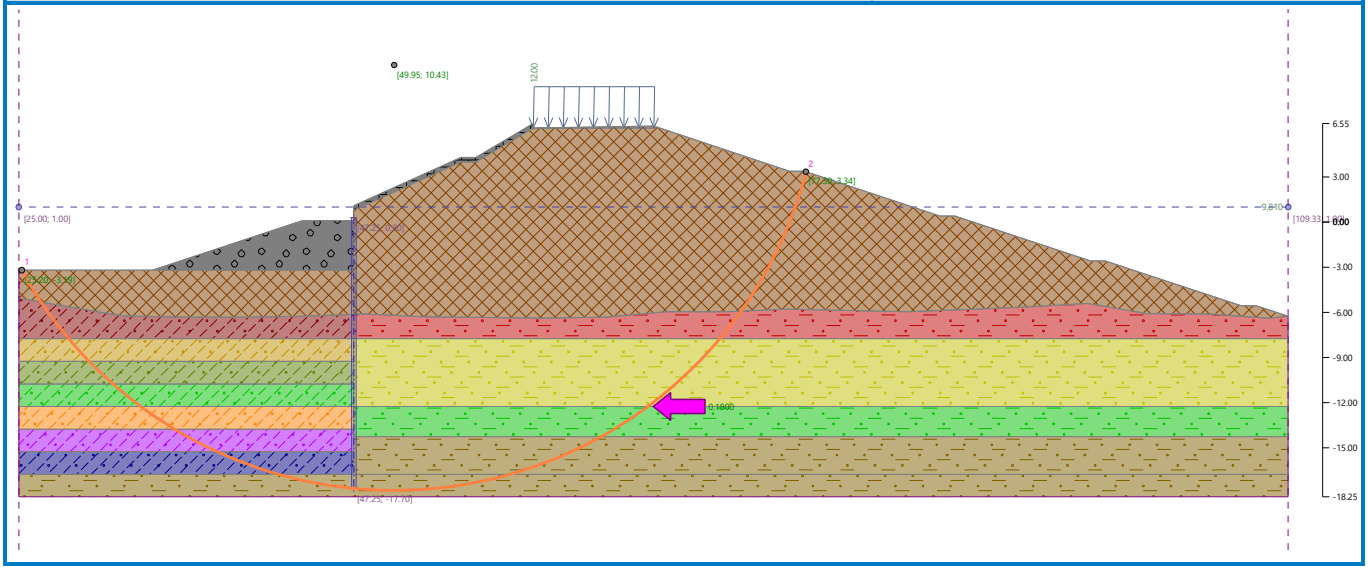
Slope stability verification (Bishop)

Sum of active forces : F_a = 3342.78 kN/m
Sum of passive forces : F_p = 4071.58 kN/m

Sliding moment : M_a = 94433.56 kNm/m
Resisting moment : M_p = 115022.16 kNm/m
Factor of safety = 1.22 > 1.10
Slope stability ACCEPTABLE

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Name : Global Slope Stabilization Analysis	Stage - analysis : 4 - 1



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Project Name:						Engineer In-Charge:	RDL
Section Name:	**TRDS RB**	Station No.:	**STA. 3+750**	Borehole No.:	**BH-86**	Checked By:	JCO

TRDS RB STA. 3+750 (BH-86) HIGHEST EMBANKMENT

SOIL LAYER PROPERTIES

Soil Layer	General					Mechanical				Groundwater			Interfaces	GWL	DFL
	Classification	Height	Width	Unit Weight	Void Ratio	Young's Modulus	Poisson's Ratio	Cohesion	Angle of Friction	Classification Type	Soil Class	Void Ratio Dependency	Cross Permeability		
		H, m	*W, m*	*δ, kN/m³*	*e*	*E, kPa*	*v*	*c, kPa*	*ϕ, °*					*m*	*m*
1	SM1	1.5	80	19	0.9	30000	0.3	10	35	USDA	Silt	☑	Fully Permeable	1	11.047
2	MH1	4.5	80	17	1	20000	0.3	79.39	5	USDA	Silty Clay	☑	Fully Permeable		
3	MH2	2	80	17	1.1	20000	0.3	69.53	5	USDA	Silty Clay	☑	Fully Permeable		
4	SM2	10	80	16.92	0.72	4667	0.187	1	7.889	USDA	Silt	☑	Fully Permeable		

EMBANKMENT PROPERTIES

Soil Layer	General					Mechanical				Groundwater			Interfaces	CAD Measurements	
	Classification	Height	Base Width	Unit Weight	Void Ratio	Young's Modulus	Poisson's Ratio	Cohesion	Angle of Friction	Classification Type	Soil Class	Void Ratio Dependency	Cross Permeability	Embankment Elevation	Base Width
		H, m	*W, m*	*δ, kN/m³*	*e*	*E, kPa*	*v*	*c, kPa*	*ϕ, °*					*m*	*W, m*
1	EMBANKMENT	13.5	79.5	20	-	50000	0.3	10	35	USDA	Sand	☑	Fully Permeable	12.65	61.63

Note: *Embankment is divided into layers, every 2m layer of soil.*

STEEL SHEET PILE PROPERTIES - PLATE

WEIGHT	A	I	E	EA	EI	v	MATERIAL TYPE
kN/m/m	*m²/m*	*m⁴/m*	*kN/m²*	*kN/m*	*kN-m²/m*	*kN-m²/m*	
10	0.03172	0.00130403	200000000	**6344000**	**260806**	**0.28**	Elastic

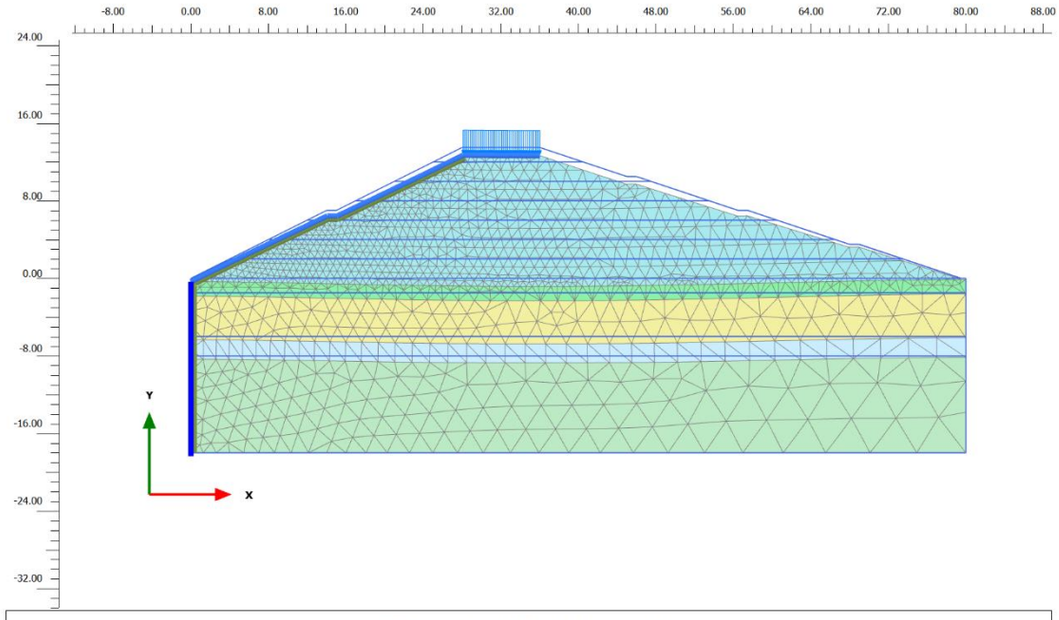
CONCRETE (REVETMENT) PROPERTIES - PLATE

WEIGHT	THICKNESS	WIDTH	fc'	E	AREA	I	EA	EI	v	MATERIAL TYPE	CAD Measurement Road Width
kN/m/m	*m*	*m*	*MPa*	*kN/m²*	*m²/m*	*m⁴/m*	*kN/m*	*kN-m²/m*	*kN-m²/m*		*m*
7.2	0.3	1	28	24870062.3	0.3	0.00225	**7461018.7**	**55957.6402**	**0.28**	Elastic	8

PILE PROPERTIES - EMBEDDED BEAM										
DIAMETER	**UNIT WEIGHT**	**E**	**AXIAL SKIN RESISTANCE**	**$T_{skin,start,max}$**	**$T_{skin,end,max}$**	**F_{max}**	**LENGTH**	**SPACING**	**NUMBER OF PILES**	**COARSENESS FACTOR**
D, m	*δ, kN/m^3*	*kN/m^2*		*kN/m*	*kN/m*	*kN*	*L, m*	*S, m*		
0.3	6	30000000	Linear	200	500	10000	-	-	-	0.7071
Note: *1st pile is 6m away from steel sheet pile and 1.5m below 1st soil layer.*										

TRDS RB STA. 3+750 (BH-86) HIGHEST EMBANKMENT										
SUMMARY OF RESULTS										
SEGMENT	**STATION**	**EMBANKMENT HEIGHT**	**NUMBER OF PILES**	**LENGTH OF PILES**	**SPACING OF PILES**	**GWL (SAFETY)**		**DFL (SAFETY)**		**Remarks**
						Deformation	**Factor of Safety**	**Deformation**	**Factor of Safety**	
		m		*m*	*m*	*m*		*m*		
TRDS RB	STA. 3+750	13.5	-	-	-	0.8579	1.365	0.626	1.517	

TRDS RB STA. 3+750 - HIGHEST EMBANKMENT

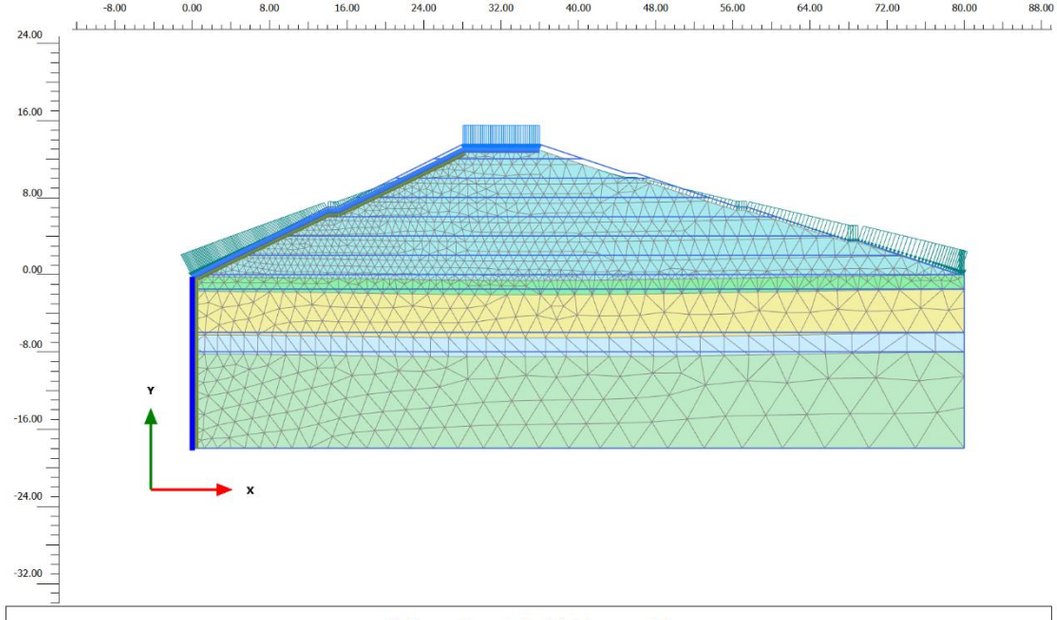


Deformed mesh |u| (at true scale)

Maximum value = 0.8579 m (Element 535 at Node 4536)

Figure 1: DEFORMATION (GWL)

TRDS RB STA. 3+750 - HIGHEST EMBANKMENT

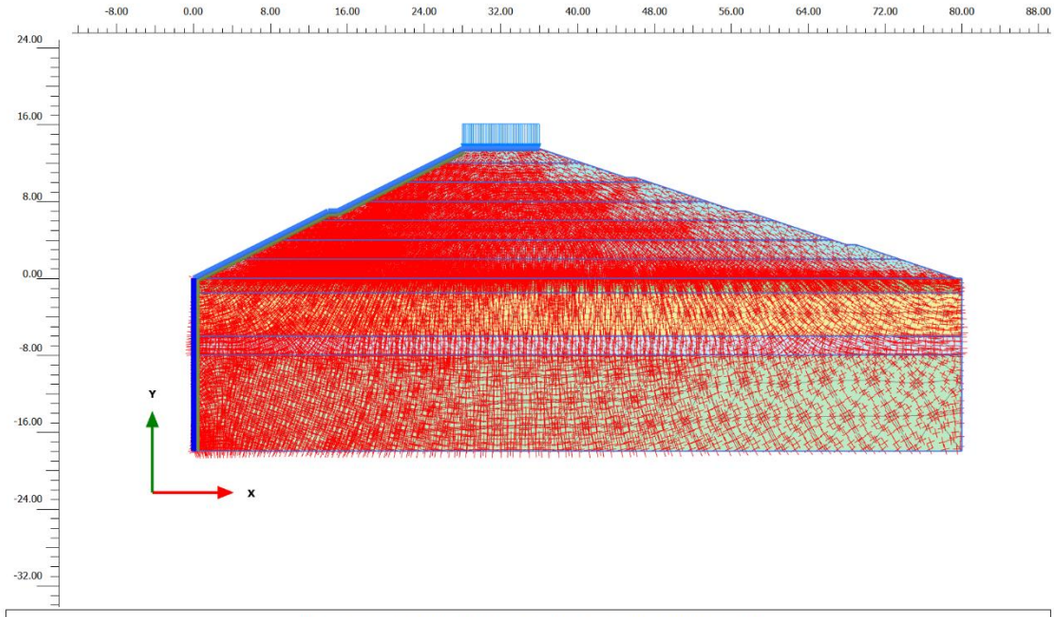


Deformed mesh |u| (at true scale)

Maximum value = 0.6260 m (Element 537 at Node 5018)

Figure 2: DEFORMATION (DFL)

TRDS RB STA. 3+750 - HIGHEST EMBANKMENT



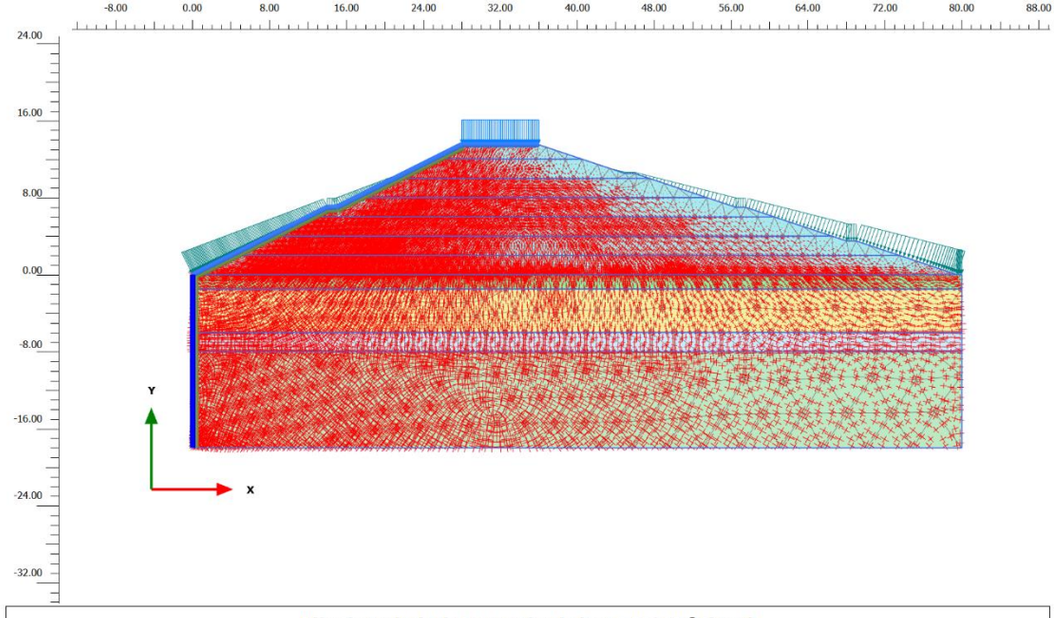
Effective principal stresses (scaled up $5.00*10^{-3}$ times)

Maximum value = $7.105*10^{-15}$ kN/m² (Element 61 at Stress point 727)

Minimum value = -361.2 kN/m² (Element 1988 at Stress point 23854)

Figure 3: EFFECTIVE STRESS DISTRIBUTION (GWL)

TRDS RB STA. 3+750 - HIGHEST EMBANKMENT



Effective principal stresses (scaled up $5.00*10^{-3}$ times)

Maximum value = $0.07105*10^{-12}$ kN/m² (Element 17 at Stress point 204)

Minimum value = -285.2 kN/m² (Element 1989 at Stress point 23868)

Figure 4: EFFECTIVE STRESS DISTRIBUTION (DFL)

TRDS RB STA. 3+750 - HIGHEST EMBANKMENT

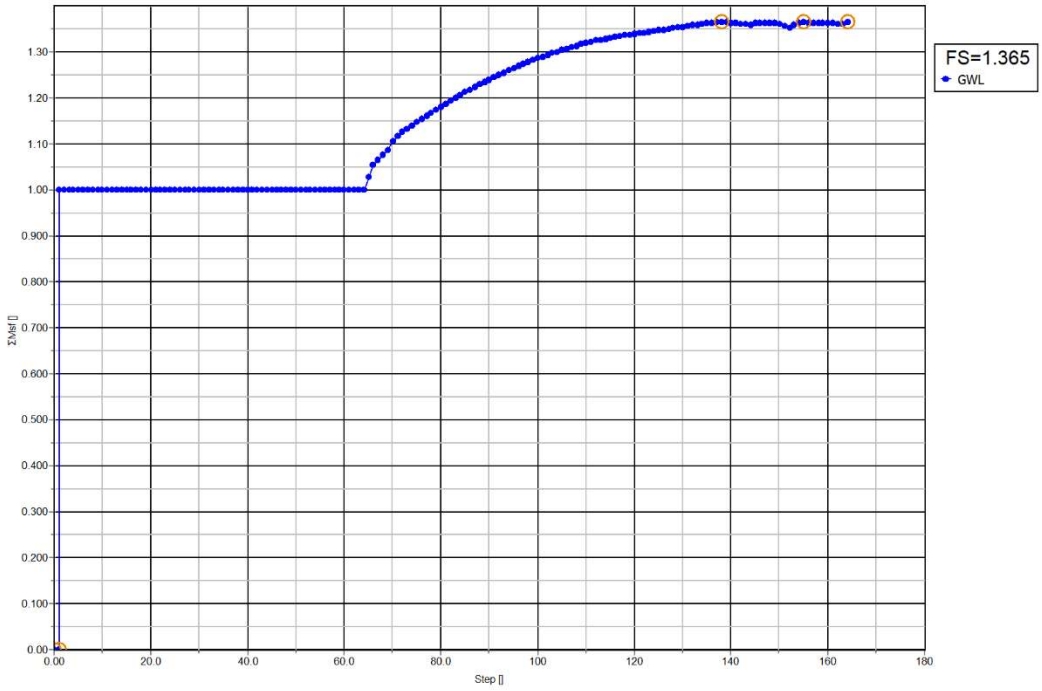


Figure 5: FACTOR OF SAFETY (GWL)

TRDS RB STA. 3+750 - HIGHEST EMBANKMENT

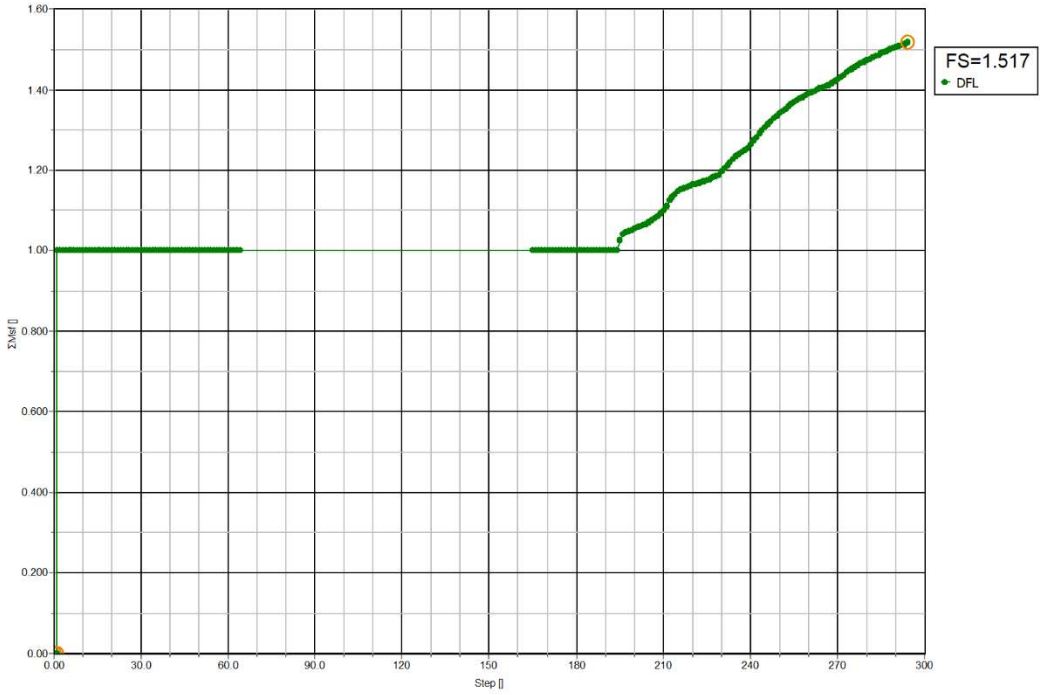


Figure 6: FACTOR OF SAFETY (DFL)